

Stationary models I

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What is time series analysis?

- Time series: data with *time ordering*.
- Observations often not i.i.d. or even serially uncorrelated. “Random sampling” from super-population not relevant.
- Classic time series analysis: few time series, focus on analyzing dependence across time, asymptotics as $T \rightarrow \infty$.
- Modern applications also: many time series (panel data with large N and T and cross-sectional dependence), high-frequency data.
- Issues that receive more focus than in cross-sectional econometrics: strong dependence across time, trends, seasonality, structural breaks, prediction/forecasting, structural Bayesian estimation.

Stochastic processes

Definition

An n -dimensional **stochastic process** is a collection $\{y_t\}_{t \in \mathcal{T}}$ of n -dim. random vectors defined on a probability space $(\Omega, \mathcal{F}, P)$.

- In this course, t denotes “time”, and often $\mathcal{T} = \mathbb{Z}$ or $\mathbb{N}$.
- For any fixed outcome $\omega \in \Omega$, the function $t \mapsto y_t(\omega)$ is called a **realization** or **sample path** of $\{y_t\}$.
- The distribution of a stochastic process is summarized by the distribution functions

$$F_{t_1, \dots, t_k}(c_1, \dots, c_k) \equiv P(y_{t_1} \leq c_1, \dots, y_{t_k} \leq c_k)$$

for all finite collections of time points $t_1, t_2, \dots, t_k \in \mathcal{T}$.

- Randomness best thought of as being over “alternative histories of the world” (different realizations of underlying shocks).

Strict stationarity

- To learn anything useful from historical data, we must make some kind of assumption that “the present is like the past”. Example:

Definition

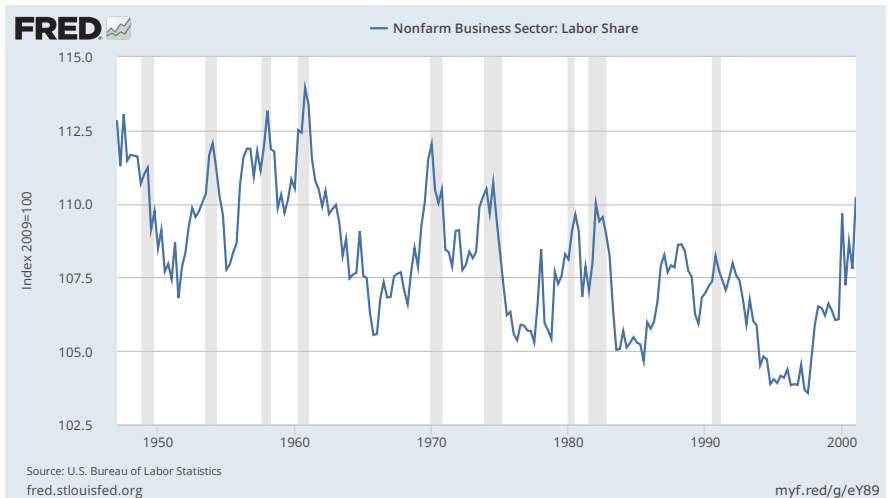
The process $\{y_t\}$ is **strictly stationary** if

$$(y_t, \dots, y_{t+k}) \stackrel{d}{=} (y_{t+\ell}, \dots, y_{t+\ell+k})$$

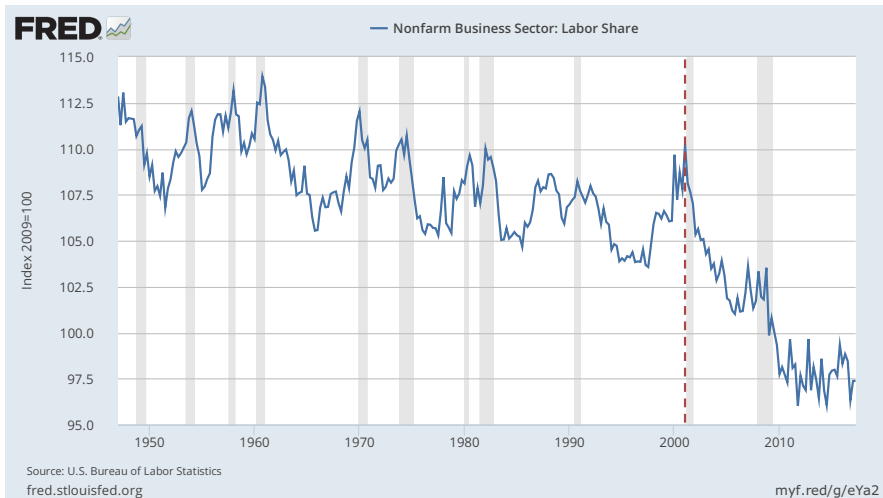
for all k, ℓ . Here “ $\stackrel{d}{=}$ ” means “has the same joint distribution as”.

- The distribution of a data subsample of any given length should not depend on which point in time the subsample starts.
- If $\{y_t\}$ is strictly stationary, then so is $\{f(y_t)\}$ for measurable $f(\cdot)$.
- Stationarity does not rule out serial corr., volatility clustering, etc.

Example: U.S. labor share, 1947q1–2000q4



Example: U.S. labor share, 1947q1–2017q2



Covariance stationarity

- In econometrics, it is common to work with an assumption that is weaker than strict stationarity.

Definition

The process $\{y_t\}$ is **covariance stationary** if *i*) $E(y_{it}^2) < \infty$ for all i and t , and *ii*) neither $E(y_t)$ nor

$$\text{Cov}(y_t, y_{t-\ell}) \equiv E[(y_t - E(y_t))(y_{t-\ell} - E(y_{t-\ell}))']$$

depend on t .

- Places the focus on the second-moment properties of the data.
- Given assumption, makes sense to define

$$\mu_y = E(y_t), \quad \Gamma_y(\ell) = \text{Cov}(y_t, y_{t-\ell}), \quad \ell \in \mathbb{Z}.$$

Autocovariance function

- The **autocovariance (matrix) function**

$$\Gamma_y(\ell) = \begin{pmatrix} \text{Cov}(y_{1t}, y_{1,t-\ell}) & \dots & \text{Cov}(y_{1t}, y_{n,t-\ell}) \\ \vdots & \ddots & \vdots \\ \text{Cov}(y_{nt}, y_{1,t-\ell}) & \dots & \text{Cov}(y_{nt}, y_{n,t-\ell}) \end{pmatrix}$$

has the following properties under cov. stationarity.

- $\Gamma_y(\ell) = \Gamma_y(-\ell)'$. In particular, $\Gamma_y(0)$ is symmetric.
- $|\Gamma_{y,ij}(\ell)| \leq \sqrt{\Gamma_{y,ii}(0)\Gamma_{y,jj}(0)}$. In particular, $|\Gamma_{y,ii}(\ell)| \leq \Gamma_{y,ii}(0)$.
- Positive semidefiniteness:** For all $k \geq 1$ and all constant vectors $a_1, \dots, a_k \in \mathbb{R}^n$,

$$\sum_{\ell=1}^k \sum_{m=1}^k a_\ell' \Gamma_y(\ell - m) a_m \geq 0.$$

- Exercise:** Verify these properties.

Autocorrelation function

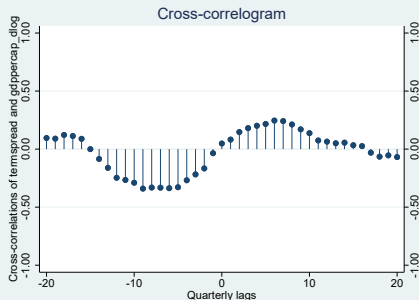
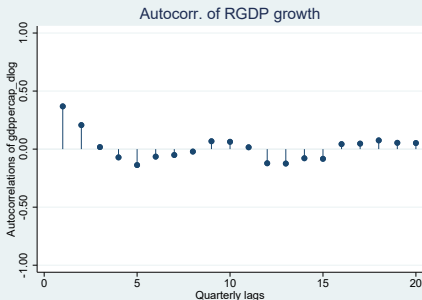
- We can also define the **autocorrelation function**

$$R_y(\ell) = \begin{pmatrix} \text{Corr}(y_{1t}, y_{1,t-\ell}) & \dots & \text{Corr}(y_{1t}, y_{nt,t-\ell}) \\ \vdots & \ddots & \vdots \\ \text{Corr}(y_{nt}, y_{1,t-\ell}) & \dots & \text{Corr}(y_{nt}, y_{nt,t-\ell}) \end{pmatrix},$$

$$\text{i.e., } R_{y,ij}(\ell) = \frac{\Gamma_{y,ij}(\ell)}{\sqrt{\Gamma_{y,ii}(0)\Gamma_{y,jj}(0)}}.$$

- It has the same properties as the ACF on the previous slide, and additionally $R_{y,ii}(0) = 1$ for all i .
- The sample analogue is called the **(cross-)correlogram**. It is useful for gauging serial correlation of individual time series, as well as lead/lag relationships between different series.

Example: Cross-corr. of GDP growth and term spread

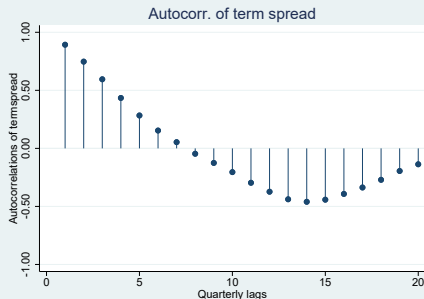


$$y_t = \begin{pmatrix} \Delta \log(\text{RGDP}_t) \\ \text{spread}_t \end{pmatrix}$$

RGDP: per capita, seas. adj.

Spread: 10yr-3mth Treasury

Sample: U.S. 1982q1-2017q2



Best linear predictor

Definition

The **best linear predictor** (BLP) of scalar random variable Y given the n -dimensional random vector X is given by

$$E^*(Y | X) \equiv \beta' X, \quad \text{where}$$

$$\beta \equiv \operatorname{argmin}_{b \in \mathbb{R}^n} E[(Y - b'X)^2].$$

We implicitly require $E(Y^2) < \infty$ and $E(X_i^2) < \infty$ for all i .

- The best linear predictor is (almost surely) unique.
- If $E(XX')$ is nonsingular, then $\beta = E(XX')^{-1}E(XY)$.

Least squares projection

- The BLP is often called the **least squares projection**. Can also be defined for projection onto an *infinite* no. of variables using Hilbert space theory. See ch. 2 of Brockwell & Davis (1991).
- Let $\{X_i\}_{i \in \mathcal{I}}$ be a countable collection of scalar r.v.'s. The **closed linear span** $\overline{\text{sp}}(X_i, i \in \mathcal{I})$ is the smallest closed linear subspace containing all $\{X_i\}_{i \in \mathcal{I}}$ (= space of all limits of sums of X_i 's).
- The *projection theorem* (B&D Thm. 2.3.1) states that there exists a unique r.v. $\hat{Y} \in \overline{\text{sp}}(X_i, i \in \mathcal{I})$ s.t.

$$E[(Y - \hat{Y})^2] = \inf_{Z \in \overline{\text{sp}}(X_i, i \in \mathcal{I})} E[(Y - Z)^2].$$

It satisfies the orthogonality conditions

$$E[(Y - \hat{Y})\tilde{X}] = 0 \quad \text{for all } \tilde{X} \in \overline{\text{sp}}(X_i, i \in \mathcal{I}).$$

- I will write $\hat{Y} \equiv E^*(Y \mid \{X_i\}_{i \in \mathcal{I}})$.

White noise

Definition

An n -dim. cov. stationary process $\{y_t\}$ is **white noise** if $\mu_y = 0_{n \times 1}$, $\Gamma_y(0) = \Sigma$, and $\Gamma_y(\ell) = 0_{n \times n}$ for all $\ell \neq 0$. We then write $y_t \sim WN(0, \Sigma)$.

- A white noise process is serially uncorrelated and so *linearly* unpredictable based on its *own* lags:

$$E^*(y_{it} \mid \{y_\tau\}_{-\infty < \tau < t}) = E(y_{it}) = 0 \quad \text{for all } i.$$

- Sufficient condition: $E(y_t \mid \{y_\tau\}_{-\infty < \tau < t}) = 0_{n \times 1}$. Then we say that y_t is a **martingale difference sequence** (MDS) wrt. the natural filtration (the sigma algebra generated by lags of y_t).
 - ▶ Example: Rational expectation forecast error.
- A white noise process may be *nonlinearly* predictable.
 - ▶ **Exercise:** Show $y_t = \cos(\omega t)$ is white noise, where $\omega \sim \text{Uniform}(0, 2\pi)$.

Gaussian processes

- An n -dim. **Gaussian process** y_t is a cov. stat. process for which $(y'_t, y'_{t+1}, \dots, y'_{t+k})'$ has a multivariate normal distr., for all t, k .
- Important properties:
 - ▶ Distribution is completely characterized by mean μ_y and ACF $\Gamma_y(\cdot)$. No additional information in higher moments.
 - ▶ Strictly stationary.
 - ▶ Any linear function of $y_t, y_{t-1}, \dots$ is also a Gaussian process.
 - ▶ Best linear predictors equal conditional expectations, e.g.:

$$E^*(y_{it} \mid 1, \{y_\tau\}_{-\infty < \tau < t}) = E(y_{it} \mid \{y_\tau\}_{-\infty < \tau < t}).$$

- ▶ If $y_t \sim WN(0, \Sigma)$ and y_t is a Gaussian process, then $y_t \stackrel{iid}{\sim} N(0, \Sigma)$.

Vector Moving Average

Definition

Let $\mu \in \mathbb{R}^n$, let Σ and Θ_ℓ ($1 \leq \ell \leq q$) be $n \times n$ matrices, and set $\Theta_0 = I_n$. The process

$$y_t = \mu + \sum_{\ell=0}^q \Theta_\ell u_{t-\ell}, \quad u_t \sim WN(0, \Sigma),$$

is called a **Vector Moving Average** of order q , $VMA(q)$.

- **Exercise:** Derive its ACF

$$\Gamma_y(\ell) = \begin{cases} \sum_{m=0}^{q-\ell} \Theta_m \Sigma \Theta'_{m+\ell}, & 0 \leq \ell \leq q, \\ 0_{n \times n}, & \ell > q. \end{cases}$$

- The elements of $\Theta_\ell = (\Theta_{ij,\ell})_{ij}$ are called **impulse responses**, since

$$E^*(y_{i,t+\ell} \mid u_t) - E(y_{i,t+\ell}) = \sum_{j=1}^n \Theta_{ij,\ell} u_{jt}, \quad 0 \leq \ell \leq q.$$

- VMA processes show up all the time in structural macro, as we'll see.

Linear filters

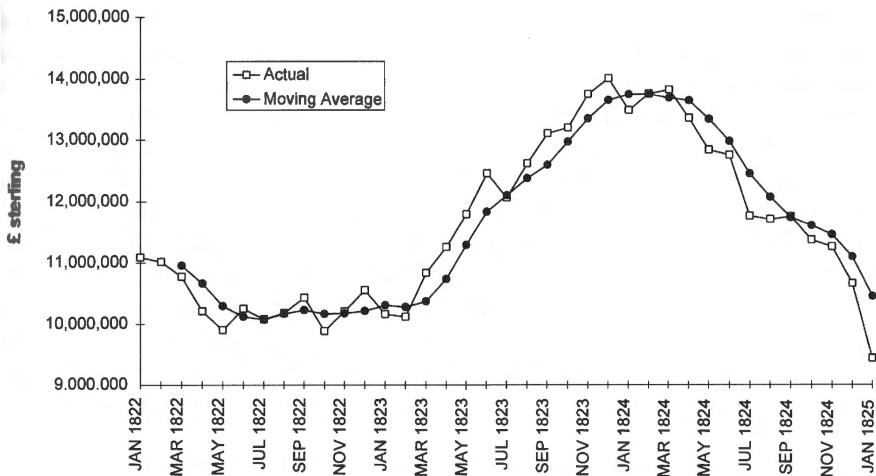
- VMA processes are defined by taking linear combinations of white noise processes. More generally, we can take linear combinations of any time series and their lags.
- If x_t is n_x -dimensional and cov. stationary, and the matrices Ψ_ℓ ($\ell \in \mathbb{Z}$) are $n_y \times n_x$, then the n_y -dimensional process

$$y_t \equiv \sum_{\ell=-\infty}^{\infty} \Psi_\ell x_{t-\ell}$$

is called a **linear filter** of $\{x_t\}$.

- Examples:
 - ▶ First difference: $y_t = \Delta x_t \equiv x_t - x_{t-1}$.
 - ▶ Centered moving average: $y_t = \frac{1}{2q+1} \sum_{\ell=-q}^q x_{t-\ell}$.
 - ▶ Exponential smoothing: $y_t = (1 - e^{-\alpha}) \sum_{\ell=0}^{\infty} e^{-\alpha \ell} x_{t-\ell}$.

Example: Bank of England's bullion stock 1822–1825



3-month trailing moving average of value of gold bullion holdings.

Source: J. L. Klein (1997), "Statistical Visions in Time".

Existence of linear filter

Proposition (Brockwell & Davis, 1991, Prop. 3.1.2)

Let $\{x_t\}$ be a covariance stationary n_x -dimensional time series. Let the matrices $\Psi_\ell = (\Psi_{ij,\ell})_{ij}$ be $n_y \times n_x$ for every $\ell \in \mathbb{Z}$, with **absolutely summable** components across ℓ :

$$\sum_{\ell=-\infty}^{\infty} |\Psi_{ij,\ell}| < \infty \quad \text{for all } i, j.$$

Then the series $\sum_{\ell=-\infty}^{\infty} \Psi_\ell x_{t-\ell}$ converges with probability 1 for every $t \in \mathbb{Z}$. Moreover, the n_y -dimensional time series $y_t = \sum_{\ell=-\infty}^{\infty} \Psi_\ell x_{t-\ell}$ is covariance stationary with ACF

$$\Gamma_y(\ell) = \sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \Psi_k \Gamma_x(\ell + m - k) \Psi'_m, \quad \ell \in \mathbb{Z}.$$

- **Exercise:** Verify formula for $\Gamma_y(\cdot)$.

Lag operator

- If $\{x_t\}$ is a stoch. process, the **lag operator** L is defined such that

$$Lx_t = x_{t-1} \quad \text{for all } t.$$

- The lag operator has the following properties:
 - ▶ L is a linear operator.
 - ▶ L^{-1} exists and is given by $L^{-1}x_t = x_{t+1}$ (the **lead operator**).
 - ▶ For any $d \in \mathbb{Z}$, $L^d x_t \equiv L(L(\dots(Lx_t))\dots) = x_{t-d}$, and L^d is a linear operator. In particular, L^0 is the identity: $L^0 x_t = x_t$.
- Some authors write B instead of L .

Lag polynomials

- If $\Psi(z) = \sum_{\ell=-\infty}^{\infty} \Psi_{\ell} z^{\ell}$ is a matrix polynomial in the scalar z , and Ψ_{ℓ} is absolutely summable across ℓ , the proposition on the previous slide implies that it makes sense to define the **lag polynomial**

$$\Psi(L) \equiv \sum_{\ell=-\infty}^{\infty} \Psi_{\ell} L^{\ell}.$$

- This means no more or less than

$$\Psi(L)x_t = \sum_{\ell=-\infty}^{\infty} \Psi_{\ell} x_{t-\ell}$$

for any cov. stationary process $\{x_t\}$.

- Terminology:
 - ▶ $\Psi(L) = \sum_{\ell=-\infty}^{\infty} \Psi_{\ell} L^{\ell}$ is **two-sided**.
 - ▶ $\Psi(L) = \sum_{\ell=0}^{\infty} \Psi_{\ell} L^{\ell}$ is **one-sided**.

Lag polynomials (cont.)

- The linearity of the lag operator means that lag polynomials have the same properties as regular polynomials, *as long as* we only use absolutely summable polynomials.
 - ▶ If $\Psi(L)$ and $\Lambda(L)$ are both absolutely summable, with conformable dimensions, then $\zeta(L) \equiv \Psi(L)\Lambda(L)$ is an abs. summable lag polynomial.
 - ▶ $\zeta(L) \equiv \Psi(L)\Lambda(L) = \sum_{\ell=-\infty}^{\infty} \zeta_{\ell} L^{\ell}$, where $\zeta_{\ell} = \sum_{m=-\infty}^{\infty} \Psi_m \Lambda_{\ell-m}$.
 - ▶ If either $\Psi(L)$ or $\Lambda(L)$ is scalar, then $\Psi(L)\Lambda(L) = \Lambda(L)\Psi(L)$.
 - ▶ The product of two one-sided lag polynomials is itself one-sided.
 - ▶ If c is a constant vector, $\Psi(L)c = \Psi(1)c = (\sum_{\ell=-\infty}^{\infty} \Psi_{\ell})c$.
- We often use the notation $\Psi(z)$ and $\Psi(L)$ interchangeably to denote the regular polynomial or lag polynomial, depending on context.
- Working with lag poly's that are *not* abs. summable is dangerous. . .

Vector Autoregression

Definition

Let $\nu \in \mathbb{R}^n$, let Σ and A_ℓ ($1 \leq \ell \leq p$) be $n \times n$ matrices, and let $u_t \sim WN(0, \Sigma)$. A covariance stationary process $\{y_t\}$ satisfying

$$y_t = \nu + \sum_{\ell=1}^p A_\ell y_{t-\ell} + u_t \quad \text{for all } t \in \mathbb{Z}$$

is called a **Vector Autoregression** of order p , $\text{VAR}(p)$.

- Defining $A(L) = I_n - \sum_{\ell=1}^p A_\ell L^\ell$, we can also write

$$A(L)y_t = \nu + u_t.$$

- VARs are ubiquitous in forecasting and structural macro.
- A cov. stationary process satisfying the above equation may not exist, depending on the A_ℓ 's. Next slides: When does it exist?

Invertible lag polynomials

- Suppose we could find an $n \times n$ abs. summ., one-sided lag polynomial $\Psi(L) = \sum_{\ell=0}^{\infty} \psi_{\ell} L^{\ell}$ such that $\Psi(L)A(L) = I_n$. Then the VAR equation

$$A(L)y_t = \nu + u_t$$

implies that y_t follows the cov. stationary VMA(∞) process

$$y_t = \Psi(L)A(L)y_t = \mu + \Psi(L)u_t, \quad \mu \equiv \Psi(1)\nu.$$

- A $\Psi(L)$ satisfying the above conditions is called a **one-sided inverse** of $A(L)$ and we write

$$\Psi(L) = A(L)^{-1}.$$

- Exercise:** For $\rho \in (-1, 1)$, show that

$$(1 - \rho L)^{-1} = \sum_{\ell=0}^{\infty} \rho^{\ell} L^{\ell}.$$

Invertible lag polynomials (cont.)

Proposition (Brockwell & Davis, 1991, Thm. 11.3.1)

Let $A(L) = \sum_{\ell=0}^p A_{\ell} L^{\ell}$. If the polynomial $z \mapsto \det(A(z))$ in the complex scalar z has all its roots (zeros) outside the unit circle ($|z| > 1$), then there exists an abs. summable, one-sided $A(L)^{-1}$.

- Proof sketch for the univariate case ($n = 1$) in the slide appendix.
- Example: VAR(1) model, $A(L) = I_n - A_1 L$.
 - ▶ $\det(A(z)) = 0$ iff z^{-1} is an eigenvalue of A_1 .
 - ▶ Abs. summable, one-sided $A(L)^{-1}$ exists if all eigenvalues of A_1 are inside the unit circle.
- For brevity, we say the “roots of $A(L)$ ”, even though they’re really the roots of $z \mapsto \det(A(z))$.

VAR stationarity

- We have seen that a cov. stationary VAR process exists if all roots of $A(L)$ lie outside the unit circle. How to compute roots in practice?
- We can write a $\text{VAR}(p)$ in **companion form**, i.e., as a large $\text{VAR}(1)$:

$$\tilde{y}_t = \tilde{\nu} + \tilde{A}\tilde{y}_{t-1} + \tilde{u}_t,$$

where

$$\tilde{y}_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix}, \quad \tilde{u}_t = \begin{pmatrix} u_t \\ 0_{n \times 1} \\ \vdots \\ 0_{n \times 1} \end{pmatrix}, \quad \tilde{\nu} = \begin{pmatrix} \nu \\ 0_{n \times 1} \\ \vdots \\ 0_{n \times 1} \end{pmatrix}, \quad \tilde{A} = \left(\begin{array}{ccc|c} A_1 & \cdots & A_{p-1} & A_p \\ \hline & I_{n(p-1)} & & 0_{n(p-1) \times n} \end{array} \right).$$

- Using $\text{VAR}(1)$ result: The $\text{VAR}(p)$ has a cov. stationary solution if all eigenvalues of the **companion matrix** $\tilde{A}$ are strictly inside the unit circle.

VAR impulse responses

- We saw that if the $\text{VAR}(p)$ stationarity condition holds, then a cov. stat. VAR process exists and it has the $\text{VMA}(\infty)$ representation:

$$y_t = \mu + \Psi(L)u_t, \quad \Psi(L) = A(L)^{-1}, \quad \mu = A(1)^{-1}\nu.$$

- Hence, we can define impulse responses $\Psi_{ij,\ell}$ as in VMA models.
- VAR impulse responses can in fact be computed by recursion:

$$\Psi_0 = I_n,$$

$$\Psi_\ell = \sum_{m=1}^{\min\{\ell, p\}} A_m \Psi_{\ell-m}, \quad \ell = 1, 2, \dots$$

- The $\text{VAR}(p)$ stat. condition from the previous slide is precisely what is needed to ensure that impulse responses don't explode as $\ell \rightarrow \infty$. Hence, the stationarity cond'n is also called the **stability** cond'n.

VAR ACF

- The VAR equation

$$y_t = \nu + \sum_{\ell=1}^p A_{\ell} y_{t-\ell} + u_t$$

implies (under cov. stationarity) the **Yule-Walker equations**

$$\Gamma_y(\ell) = \begin{cases} \sum_{m=1}^p A_m \Gamma_y(m)' + \Sigma, & \ell = 0, \\ \sum_{m=1}^p A_m \Gamma_y(\ell - m), & \ell \geq 1. \end{cases}$$

- Can solve for $\Gamma_y(0), \Gamma_y(1), \dots, \Gamma_y(p)$ using the eq'ns for $\ell = 0, 1, \dots, p$ and $\Gamma_y(-\ell) = \Gamma_y(\ell)'$. Then $\Gamma_y(p+1), \dots$ follow.
- **Exercise:** Show that in the VAR(1) model,

$$\text{vec}(\Gamma_y(0)) = (I_{n^2} - A_1 \otimes A_1)^{-1} \text{vec}(\Sigma),$$

$$\Gamma_y(\ell) = A_1^{\ell} \Gamma_y(0), \quad \ell \geq 1.$$

Can transform a VAR(p) to a VAR(1) using the companion form.

VARMA models

Definition

Let $\mu \in \mathbb{R}^n$, and let Σ , A_ℓ ($1 \leq \ell \leq p$), and Θ_ℓ ($1 \leq \ell \leq q$) be $n \times n$ matrices. Let $u_t \sim WN(0, \Sigma)$. Set $A(L) = I_n - \sum_{\ell=1}^p A_\ell L^\ell$, $\Theta(L) = I_n + \sum_{\ell=1}^q \Theta_\ell L^\ell$. A cov. stat. process $\{y_t\}$ satisfying

$$A(L)(y_t - \mu) = \Theta(L)u_t \quad \text{for all } t \in \mathbb{Z},$$

is called a **VARMA**(p, q) process.

- VARMAs generalize VARs and VMAs, but in practice we often deal with either VARs or VMAs.

Causal and invertible processes

$$A(L)y_t = \Theta(L)u_t.$$

- The VARMA process $\{y_t\}$ is said to be **causal** wrt. $\{u_t\}$ if

$$y_t \in \overline{\text{sp}}(u_\tau, -\infty < \tau \leq t).$$

- ▶ That is, y_t driven only by *past* and *current* shocks u_t .
- ▶ Sufficient condition: one-sided inverse $A(L)^{-1}$ exists. Yields VMA(∞) representation $y_t = \Psi(L)u_t$, where $\Psi(L) = \sum_{\ell=0}^{\infty} \Psi_\ell L^\ell \equiv A(L)^{-1}\Theta(L)$.

- The VARMA process $\{y_t\}$ is said to be **invertible** wrt. $\{u_t\}$ if

$$u_t \in \overline{\text{sp}}(y_\tau, -\infty < \tau \leq t).$$

- ▶ That is, we can recover shocks from past and current y_t .
- ▶ Sufficient condition: *one-sided* inverse $\Theta(L)^{-1}$ exists.

Prediction/forecasting

- Consider time series $\{y_t\}$ with *causal, invertible* VMA representation

$$y_t = \Theta(L)u_t, \quad u_t \sim WN(0, \Sigma).$$

(Mean zero for notational simplicity.)

- Then the ℓ -step-ahead least-squares-optimal forecast is given by

$$\begin{aligned} E^*(y_{i,t+\ell} \mid \{y_\tau\}_{-\infty < \tau \leq t}) &= E^*(y_{i,t+\ell} \mid \{u_\tau\}_{-\infty < \tau \leq t}) \\ &= \sum_{m=0}^{\infty} e_i' \Theta_{m+\ell} u_{t-m}, \end{aligned}$$

where $u_t = \Theta(L)^{-1}y_t$.

- Explicit VARMA forecast formulas in B&D ch. 5.2–5.3, 11.4.
- For VAR(1) models, the formula reduces to the simple expression

$$E^*(y_{i,t+\ell} \mid \{y_\tau\}_{-\infty < \tau \leq t}) = E^*(y_{i,t+\ell} \mid y_t) = e_i' A_1^\ell y_t.$$

Generalizes to VAR(p) using the companion form.

Granger-Sims causality

- Define $\text{Var}^*(Y | X) \equiv E[(Y - E^*(Y | X))^2]$ for scalar Y .

Definition

Let $\{y_t\}$ and $\{x_t\}$ be two scalar and jointly cov. stat. time series.
Then $\{x_t\}$ **Granger-causes** $\{y_t\}$ if

$$\text{Var}^*(y_{t+1} | \{x_\tau, y_\tau\}_{-\infty < \tau \leq t}) < \text{Var}^*(y_{t+1} | \{y_\tau\}_{-\infty < \tau \leq t}).$$

- In words, $\{x_t\}$ helps (linearly) predict $\{y_t\}$.
- This is about forecasting, not counterfactual causality. “The weather forecast Granger-causes the weather” (Cochrane, 1994).
- Sims (1972) showed that $\{x_t\}$ Granger-causes $\{y_t\}$ if and only if

$$\text{Var}^*(x_t | \{y_\tau\}_{-\infty < \tau < \infty}) < \text{Var}^*(x_t | \{y_\tau\}_{-\infty < \tau \leq t}),$$

i.e., future y_t 's help in predicting current x_t .

Granger-Sims causality in VARs

- Suppose $(y_t, x_t)'$ follow a bivariate VAR:

$$\begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} \nu_y \\ \nu_x \end{pmatrix} + \sum_{\ell=1}^p \begin{pmatrix} A_{yy,\ell} & A_{yx,\ell} \\ A_{xy,\ell} & A_{xx,\ell} \end{pmatrix} \begin{pmatrix} y_{t-\ell} \\ x_{t-\ell} \end{pmatrix} + \begin{pmatrix} u_{yt} \\ u_{xt} \end{pmatrix},$$

where $(u_{yt}, u_{xt})' \sim WN(0, \Sigma)$, with Σ positive definite.

- Then $\{x_t\}$ does *not* Granger-cause $\{y_t\}$ if and only if

$$A_{yx,\ell} = 0 \quad \text{for all } \ell = 1, \dots, p.$$

Simple parametric exclusion restriction, easy to test in data.

- This result can easily be extended to allow for third variables, i.e., conditioning throughout on some additional “background” explanatory variables w_t in the definition of Granger causality.

Application: What predicts output and inflation?

SUMMARY OF GRANGER CAUSALITY AND QLR TEST STATISTICS

A. Summarized by Predictor Category										
Predictor Category	Inflation					Output				
	GC	QLR	G&Q	G×Q	N	GC	QLR	G&Q	G×Q	N
Asset Prices	0.35	0.78	0.29	0.28	420	0.35	0.71	0.28	0.25	420
Activity	0.63	0.78	0.48	0.49	106	0.56	0.46	0.28	0.26	134
G&C Prices	0.32	0.79	0.25	0.26	216	0.56	0.77	0.47	0.43	188
Money	0.52	0.68	0.38	0.35	114	0.40	0.53	0.30	0.21	114
All	0.40	0.77	0.32	0.31	856	0.44	0.66	0.32	0.29	856

B. Summarized by Country										
Country	Inflation					Output				
	GC	QLR	G&Q	G×Q	N	GC	QLR	G&Q	G×Q	N
Canada	0.52	0.74	0.40	0.38	124	0.50	0.68	0.33	0.34	124
France	0.41	0.84	0.35	0.34	108	0.52	0.68	0.38	0.35	108
Germany	0.43	0.70	0.32	0.30	118	0.35	0.67	0.25	0.23	118
Italy	0.33	0.83	0.25	0.27	126	0.35	0.62	0.25	0.22	126
Japan	0.34	0.85	0.29	0.29	122	0.41	0.66	0.31	0.27	122
United Kingdom	0.33	0.54	0.21	0.18	112	0.36	0.58	0.24	0.21	112
United States	0.4	0.82	0.38	0.37	146	0.56	0.73	0.47	0.41	146
All	0.40	0.77	0.32	0.31	856	0.44	0.66	0.32	0.29	856

Notes: The Granger causality and QLR test statistics are heteroskedasticity-robust and were computed for a one-quarter ahead bivariate in-sample (full-sample) regression ($h = 1$ in equation (3)). The five numbers in each cell are the fraction of bivariate models with significant (5%) GC statistics (column label GC), significant (5%) QLR statistics (column label QLR), significant GC and QLR statistics (column label G&Q), the product of the first and second (column label $G \times Q$), and the number of models in each cell. The models making up each cell are the pooled results using CPI, the GDP deflator, real GDP, and IP.

Source: Stock & Watson (2003)

Wold decomposition

Proposition (Hannan, 1970, Theorem 2'', p. 158)

Let $\{y_t\}$ be an n -dim. cov. stationary time series. There exists an $n \times n$ lag polyn. $\Theta(L)$ and n -dim. $u_t \sim WN(0, \Sigma)$ such that

$$y_t = \Theta(L)u_t + d_t, \quad \Theta(L) = I_n + \sum_{\ell=1}^{\infty} \Theta_{\ell} L^{\ell},$$

and the following properties hold:

- ① $\sum_{\ell=1}^{\infty} \Theta_{ij,\ell}^2 < \infty$ for all (i, j) .
- ② $\{y_t\}$ is invertible wrt. $\{u_t\}$: $u_t \in \overline{\text{sp}}(y_{\tau}, -\infty < \tau \leq t)$.
- ③ $\{d_t\}$ is **deterministic**: $\text{Var}^*(d_t \mid \{d_{\tau}\}_{-\infty < \tau \leq t-1}) = 0_{n \times n}$.
- ④ $d_t \in \bigcap_{s=-\infty}^{\infty} \overline{\text{sp}}(y_{\tau}, -\infty < \tau \leq s)$ for all t .
- ⑤ $\text{Cov}(u_t, d_s) = 0_{n \times n}$ for all s, t .

Wold decomposition: interpretation

$$y_t = \Theta(L)u_t + d_t$$

- Theorem says that *any* cov. stationary process has a VMA(∞) representation (plus deterministic term d_t).
- Since $u_t \in \overline{\text{sp}}(y_\tau, -\infty < \tau \leq t)$ and $\Theta(0) = I_n$, we must have
$$u_t = y_t - E^*(y_t \mid \{y_\tau\}_{-\infty < \tau \leq t-1}), \quad \Sigma = \text{Var}^*(y_t \mid \{y_\tau\}_{-\infty < \tau \leq t-1}).$$
- So Wold decomp. splits $\{y_t\}$ into one-step-ahead prediction errors, plus a residual that is perfectly linearly predictable.
- Prediction errors u_t may have no deep economic interpretation!
- All parameters are identified: $\Theta_\ell = \text{Cov}(y_t, u_{t-\ell})\Sigma^{-1}$, $\ell \geq 1$.
- Theorem guarantees that $\Theta(L)$ is **square summable**, not abs. summable. But square summability is enough for the process to have a well-defined mean-square limit.

Wold decomposition: deterministic component

- If $d_t = 0_{n \times 1}$ in the Wold decomposition, we call $\{y_t\}$ **purely (linearly) nondeterministic**.
- Most econometric models generate purely nondeterministic time series, e.g., causal VARMA models.
- In general, however, we cannot neglect the deterministic term.
- A deterministic process is perfectly (linearly) predictable. Univariate example:

$$d_t = A \cos t + B \sin t, \quad (A, B)' \sim N(0_{2 \times 1}, I_2).$$

Easy to show that this is strictly stationary ([exercise](#): verify). Using trigonometric identities, can also show

$$d_t = 2 \cos(1) d_{t-1} - d_{t-2} \quad \text{for all } t,$$

so $\{d_t\}$ is linearly deterministic.

Appendix: Existence of one-sided inverse, proof sketch

Proposition (Brockwell & Davis, 1991, Thm. 3.1.1)

Let $\psi(L) = \sum_{\ell=0}^P \psi_{\ell} L^{\ell}$ where the ψ_{ℓ} 's are scalars. If $\psi(z) \neq 0$ for all complex z such that $|z| \leq 1$, then there exists an abs. summable one-sided inverse $\psi^{-1}(L)$.

- Since $\psi(z) \neq 0$ for all $|z| \leq 1$, and $\psi(\cdot)$ is continuous, there in fact exists $\varepsilon > 0$ such that $\psi(z) \neq 0$ for all $|z| \leq 1 + \varepsilon$.
- Thus, $\frac{1}{\psi(z)}$ has a Taylor expansion $\sum_{\ell=0}^{\infty} \theta_{\ell} z^{\ell}$ valid for all $|z| \leq 1 + \varepsilon$.
- Thus, $\psi(z)\theta(z) = 1$ for $\theta(z) = \sum_{\ell=0}^{\infty} \theta_{\ell} z^{\ell}$, as desired. *But* we still need to show the θ_{ℓ} 's are absolutely summable.
- Since $\sum_{\ell=0}^{\infty} \theta_{\ell} (1 + \varepsilon)^{\ell} < \infty$ (by validity of the Taylor expansion), we have $\theta_{\ell} (1 + \varepsilon)^{\ell} \rightarrow 0$ as $\ell \rightarrow \infty$. Thus, $|\theta_{\ell}| \leq C(1 + \varepsilon)^{-\ell}$ for some constant $C > 0$ and all large ℓ , implying abs. summability (verify).